

CoachAI - Product Backlog (AI Customer Support Assistant with Live Response Guidance)

Project: Development of AI-Powered Customer Support Assistant with Live Response Guidance | Batch: Jun-Aug 2026 | Team: MasterIJ27 + CoachAI Team | Repo: github.com/MasterIJ27 | Sprints: 6 (6 weeks) | Deadline: 28-Aug-2026

Planned Sprint	Actual Sprint	US ID	User Story / Description	MOSCOW	Dependency	Assignee	Status
1	1	US-001	As a stakeholder, I want project charter and requirement analysis covering problem (late coaching, scattered KB, late escalation) and solution (real-time copilot) so scope is clear	MUST HAVE	No Dependency	Team Lead / BA	3- Completed
1	1	US-002	As a developer, I want GitHub repo with branching, .gitignore, MIT license and README skeleton so collaboration is enabled	MUST HAVE	US-001	Team Lead	3- Completed
1	1	US-003	As an architect, I want system architecture (React + Node + Python + Postgres + LLM chain Azure -> OpenRouter -> Groq) and 9-step per-turn flow documented	MUST HAVE	US-001	AI Architect	3- Completed
1	1	US-004	As DevOps, I want env config (.env.example, DATABASE_URL, AZURE/GROQ/OPENROUTER keys, COMPOSIO keys) with graceful degradation if keys missine	MUST HAVE	US-002	Backend Team	3- Completed
1	1	US-005	As PM, I want Orchestrator and ConversationManager session lifecycle (start/process/advance/end + rehydration via full_state JSON) for serverless	MUST HAVE	US-003	AI Team	3- Completed
2	2	US-006	As agent, I want Customer Simulator (LLM temp 0.8, Hinglish <=140 chars, sentiment state machine, 6 fallbacks) for realistic training	MUST HAVE	US-005	AI Team	3- Completed
2	2	US-007	As agent, I want Intent and Sentiment Agent (intent + frustration 0-1 + trend -1..1, JSON regex fallback) in <0.5s	MUST HAVE	US-005	AI Team	3- Completed
2	2	US-008	As manager, I want Escalation Monitor (weighted: frustration*0.4 + sentiment*0.2 + trend*0.15 + keywords*0.1 + length*0.1 + repetition*0.05)	MUST HAVE	US-007	AI Team	3- Completed
2	2	US-009	As agent, I want Coaching Suggestion (suggested reply + tone/clarity 1-5 + macros library + retention STAY15) before send	MUST HAVE	US-007	AI Team	3- Completed
2	2	US-010	As QA, I want Post-Interaction Summary (overall = resolution*0.5 + coaching*0.3 + clarity*0.2) and PerformanceReport	MUST HAVE	US-009	AI Team	3- Completed
3	3	US-011	As KB admin, I want RAG ingestion (.txt/.pdf/.docx/.json/.md/.csv/.html) with paragraph -> 512-char chunks and sentence re-split	MUST HAVE	US-001	AI Team	3- Completed
3	3	US-012	As agent, I want hybrid retrieval: semantic (OpenRouter nemotron 2048d cosine, JSONB cache) + BM25/keyword offline fallback	MUST HAVE	US-011	AI Team	3- Completed
3	3	US-013	As agent, I want Knowledge Recommendation (agentic RAG 1 tip) + KB-gap <0.45 -> pending FAQ draft via Auto-KB	MUST HAVE	US-012	AI Team	3- Completed
3	3	US-014	As manager, I want Deep Analysis (1 LLM call -> CSAT 1-5, churn %, viral/PR + statement, fraud+protocol, defection+offer, mind-read)	SHOULD HAVE	US-007	AI Team	3- Completed
3	3	US-015	As analyst, I want Predictive CSAT, Defection (swiggy/zepto keywords), Viral Threat, Fraud Detector agents	SHOULD HAVE	US-014	AI Team	3- Completed
4	4	US-016	As agent, I want Mind Reader, Multiverse (2-3 what-if branches + CSAT), Cognitive Load, Patience Clock, Bot Agent (Zomato menu, escalate on human/frustration>=0.7)	SHOULD HAVE	US-014	AI Team	3- Completed
4	4	US-017	As agent, I want Manager Supervisor (whisper/takeover if >=0.5), Compliance Monitor, Tone Rewriter (<10 chars -> canned apology), Auto-Pilot (one-click refund/voucher)	SHOULD HAVE	US-009	AI Team	3- Completed
4	4	US-018	As manager, I want Jira Generator (transcript -> ticket), Scenario Generator, QA Audit (21 checks, pass if overall>=0.75)	SHOULD HAVE	US-010	AI Team	3- Completed
4	4	US-019	As integrator, I want Composio (JIRA_CREATE_ISSUE, GMAIL, Slack) + mock_backend (refund Rs.500 guard, voucher) with graceful fail	COULD HAVE	US-018	Backend Team	3- Completed
4	4	US-020	As trainee, I want Survival Mode (4 tickets, grading 0.85/0.6/0.35, HP/streak/power-ups +500) and Hall of Fame/Shame (>=0.85 / <=0.45)	COULD HAVE	US-010	AI Team	3- Completed
5	5	US-021	As user, I want Node Express backend (auth register/login/guest/me, sessionStore pg rehydration, 40+ routes, rag BM25, llam chain) port 8000.	MUST HAVE	US-005	Backend Team	3- Completed
5	5	US-022	As dev, I want Python FastAPI (server.py + features.py 20 routes, _serialize_turn) and Streamlit app (src/ui/app.py 5 pages)	MUST HAVE	US-021	Backend Team	3- Completed
5	5	US-023	As agent, I want React frontend (Vite 5 + Tailwind + react-router 6, framer-motion, query) with Landing/Login/Dashboard (Signals/KB/Reply/Risk)/Setup/Reports/Analytics/Knowledge/HallOfFame/Jira	MUST HAVE	US-021	Frontend Team	3- Completed
5	5	US-024	As user, I want API integration (api.js bearer), session/start, chat/message -> coachTurn, autopilot, manager-takeover, chat/end -> email report	MUST HAVE	US-023	Frontend Team	3- Completed
5	5	US-025	As power user, I want Feature Lab (Multiverse/Tone/Compliance/Jira), TTS fallback, Survival, analytics and jsPDF export	SHOULD HAVE	US-023	Frontend Team	3- Completed
6	6	US-026	As QA, I want tests: pytest 14 + load, vitest 7, node --test, CI on push	MUST HAVE	US-022	QA Team	3- Completed
6	6	US-027	As DevOps, I want deploy: Vercel (frontend+backend), Streamlit Cloud, Neon Postgres, /health (engine/RAG/db)	MUST HAVE	US-021	DevOps	3- Completed
6	6	US-028	As stakeholder, I want docs: 10-slide PPT exactly, KT Guide + Q&A Bank 25 Qs, HTML docs, Deep Dive, CheatSheet	MUST HAVE	No Dependency	Team Lead	3- Completed
6	6	US-029	As QA, I want hardening: ephemeral FS fix (DB authoritative), two-backends-one-Vercel guard, all agents fallback, load test	MUST HAVE	US-026	QA Team	3- Completed
6	6	US-030	As PM, I want final packaging: templates filled, GitHub release final/main, 5-min demo script, submit 28-Aug-2026	MUST HAVE	US-028	Team Lead	3- Completed

Note: 30 user stories across 6 sprints (Agile Template v0.1). MoScWoW applied. All 3-Completed for 28-Aug-2026 submission. Reference files Sample_Agile.xls and Agile_Template.xlsm NOT uploaded/edited - this file is the completed submission for CoachAI (github.com/MasterIJ27).

CoachAI - Sprint Backlog (Task Breakdown, 6 Sprints)

Sprint-wise tasks with effort (hrs), owner and acceptance criteria -> maps US-IDs to executable tasks

Sprint	Task ID	US ID	Task / Activity	Assignee	Est. Hrs	Actual Hrs	Dependency	Status
1	T-01	US-001	Requirement workshop, Infosys problem analysis, MVP scope and acceptance criteria	Team Lead/BA	8	8	-	3- Completed
1	T-02	US-002	Init GitHub repo, .gitignore, branch main, MIT license, README structure	Team Lead	4	4	T-01	3- Completed
1	T-03	US-003	Draw architecture (React-Node-Python-Postgres-LLM) and 9-step flow, document in KT guide	AI Architect	6	6	T-01	3- Completed
1	T-04	US-004	Create .env.example (15 keys), config.py with 3-layer fallback (settings->env->st.secrets)	Backend Team	5	5	T-02	3- Completed
1	T-05	US-005	Code Orchestrator singleton + ConversationManager (rehydrate via full_state JSON)	AI Team	10	10	T-03	3- Completed
2	T-06	US-006	Build customer_simulator: 6 fallbacks, Hinglish mixer, sentiment machine, <=140 chars	AI Team	8	8	T-05	3- Completed
2	T-07	US-007	Build intent_sentiment: builder, temp 0.1, JSON regex, frustration/trend heuristics	AI Team	8	8	T-05	3- Completed
2	T-08	US-008	Build escalation_monitor weighted formula + 6 unit tests	AI Team	6	5	T-07	3- Completed
2	T-09	US-009	Build coaching_suggestion: macros inject, tone/clarity + LLM, STAY15 logic	AI Team	8	8	T-07	3- Completed
2	T-10	US-010	Build post_interaction_summary scoring + report JSON	AI Team	6	6	T-09	3- Completed
3	T-11	US-011	Implement ingest.py: 7 formats, para->512 chunks, sentence re-split	AI Team	8	8	T-01	3- Completed
3	T-12	US-012	Implement hybrid search: embed cosine, BM25 fallback, JSONB cache	AI Team	10	10	T-11	3- Completed
3	T-13	US-013	Build knowledge_recommendation + gap <0.45 + auto_kb draft	AI Team	7	7	T-12	3- Completed
3	T-14	US-014	Build deep_analysis 1-call (CSAT/churn/viral/PR/fraud/defection/mind-read)	AI Team	9	9	T-07	3- Completed
3	T-15	US-015	Build predictive_csat, defection, viral, fraud (keyword pre-scan -> LLM)	AI Team	8	8	T-14	3- Completed
4	T-16	US-016	Build mind_reader, multiverse (3 branches), cognitive, patience, bot menu	AI Team	10	10	T-14	3- Completed
4	T-17	US-017	Build manager_supervisor (>=0.5 no-op), compliance, tone_rewriter (<10 guard), auto_pilot	AI Team	10	10	T-09	3- Completed
4	T-18	US-018	Build jira_bug_generator, scenario_generator, qa_audit (21 checks, pass >=0.75)	AI Team	8	8	T-10	3- Completed
4	T-19	US-019	Implement mock_backend (Rs.500 guard) + composio_backend graceful fail	Backend Team	7	7	T-18	3- Completed
4	T-20	US-020	Build survival_game (4 tickets, 0.85/0.6/0.35, shield@3, +500) + hall_of_fame	AI Team	6	6	T-10	3- Completed
5	T-21	US-021	Build Node Express backend: auth, sessionStore pg rehydration, 40+ routes	Backend Team	14	14	T-05	3- Completed
5	T-22	US-022	Build Python FastAPI (server+features 20 routes, _serialize_turn) + Streamlit	Backend Team	10	10	T-21	3- Completed
5	T-23	US-023	Build React frontend (Vite+Tailwind, 10 pages, 58% code-split) dark navy glassmorphism	Frontend Team	16	16	T-21	3- Completed
5	T-24	US-024	Integrate frontend-backend: Login/guest, Setup->Dashboard 4-tab, Autopilot, takeover, end email	Frontend Team	10	10	T-23	3- Completed
5	T-25	US-025	Build Feature Lab, Analytics, Reports jsPDF, Knowledge admin tabs	Frontend Team	8	8	T-24	3- Completed
6	T-26	US-026	Write tests: pytest 14 + load, vitest 7, node --test	QA Team	8	8	T-22	3- Completed
6	T-27	US-027	Deploy: Vercel (frontend+backend), Streamlit Cloud, Neon Postgres, /health	DevOps	8	7	T-21	3- Completed
6	T-28	US-028	Create docs: PPT 10 slides exactly, KT Guide 342 lines, HTML docs, Deep Dive, QA Bank 25 Qs	Team Lead	10	10	-	3- Completed
6	T-29	US-029	Hardening: two-backends guard, ephemeral FS -> DB, all fallbacks, load test	QA Team	6	6	T-26	3- Completed
6	T-30	US-030	Package: fill Agile/Defect/UTP, GitHub push final/main, rehearse 10min demo, submit 28-Aug	Team Lead	4	4	T-28	3- Completed

CoachAI - Daily Stand-up Log (Jun-Aug 2026, 6 Sprints)

Daily 15-min stand-ups: Yesterday / Today / Blockers - Scrum Master: Team Lead | 10:00 AM IST | Google Meet + GitHub Projects

Date	Sprint	Team Member	What was done yesterday?	What will be done today?	Blockers / Impediments	Action Items
2026-07-07	Sprint 1	Team Lead	Kick-off, requirement analysis, created GitHub repo and README skeleton	Finalize architecture diagram and per-turn flow, create env.example	None	Review architecture by EOD
2026-07-08	Sprint 1	AI Team	Implemented config.py with 3-layer key fallback, graceful degradation tested	Start Orchestrator singleton and ConversationManager skeleton	Azure key not yet provisioned - using Groq fallback	Request Azure OpenAI access
2026-07-10	Sprint 1	Backend Team	Completed Orchestrator lifecycle and DB schema (sessions full, state JSON)	Wire Streamlit sidebar and setup page	None	Demo Sprint 1 review 11-Jul
2026-07-14	Sprint 2	AI Team	Built customer_simulator (Hinglish, sentiment) + intent_sentiment (JSON regex fallback)	Build escalation_monitor weighted formula + unit tests	LLM returns markdown fence - added regex extractor	Refine prompt for JSON-only
2026-07-15	Sprint 2	AI Team	Escalation monitor done, coaching_suggestion with macros inject working	Implement post_interaction_summary scoring and report JSON	Clarity score variance high - tuning heuristics	Add fallback for clarity
2026-07-18	Sprint 2	AI Team	Sprint 2 review: 5/5 US done, demo <0.5s pipeline	Begin Sprint 3: RAG ingestion multi-format	None	Sprint 3 planning
2026-07-21	Sprint 3	AI Team	Ingest .txt/.pdf via PyPDF2, 512-char chunks, 19 FAQs loaded	Build hybrid retrieval (embedding + BM25) and JSONB cache	OpenRouter 429 - keyword fallback saved demo	Add retry + fallback log
2026-07-23	Sprint 3	AI Team	Knowledge recommendation + KB-gap <0.45 done, auto-KB draft to pending/	Implement deep_analysis one-call and 4 risk agents	Deep prompt too long - compress but keep single call	Compress prompt, keep JSON contract
2026-07-25	Sprint 3	AI Team	Deep signals working, CSAT vs heuristic verified	Test RAG on 30 queries, tune 0.45 threshold	None	Sprint 3 demo 26-Jul
2026-07-28	Sprint 4	AI Team	Built mind_reader, multiverse, cognitive_load, patience_clock, bot Agent menu	Implement manager_supervisor (0.5), compliance, tone rewriter	Manager whisper too frequent - raised to 0.5	Adjust threshold, add hysteresis
2026-07-30	Sprint 4	AI Team	Tone rewriter <10 guard done, auto_pilot + Composio mock working	Build Jira generator, scenario generator, QA audit (21 checks)	Composio SDK breaking change 0.18->0.19	Pin deps to 0.18.1
2026-08-01	Sprint 4	Backend Team	Survival arcade and Hall of Fame done, power-ups tested	Sprint 4 review, start Node backend auth and sessionStore	Survival singleton shared - known limitation	Document in gotchas
2026-08-04	Sprint 5	Backend Team	Node Express backend: auth (register/login/guest), pg rehydration, 20 routes, health	Build React frontend shell (Landing/Login/Dashboard) and bearer wrapper	Two backends same Vercel project - deploy order careful	Deploy Node first, verify /api/auth/me
2026-08-06	Sprint 5	Frontend Team	Dashboard 4-tab rail (Signals/KB/Reply/Risk) wired to /api/chat/message	Implement Feature Lab, Analytics, Reports, Knowledge admin	Risk tab overflow on mobile - fixed Tailwind	Test responsive at 360px
2026-08-08	Sprint 5	Frontend Team	Feature Lab (Multiverse/Tone/Compliance/Jira) + autopilot/takeover done	End-to-end guest->Setup->Dashboard->Report->Email test	DB rehydration on cold start verified	Add activity log
2026-08-11	Sprint 6	QA Team	pytest 14 tests + vitest 7 + node --test all green locally	Deploy to Vercel prod and Streamlit Cloud, configure DATABASE_URL	Ephemeral FS caused missing reports - switched to DB	Never read from fs on Vercel
2026-08-13	Sprint 6	Team Lead	Drafted PPT 10 slides exactly, KT Guide + Deep Dive, HTML docs	Create QA Bank 25 Qs, CheatSheet, File Structure PDFs	PPT had 12 slides - trimmed to 10	Enforce 10-slide rule
2026-08-15	Sprint 6	QA Team	Load test 50 concurrent, p95 <0.8s, fallback chain reliable (Azure->Groq)	Fill Agile/Defect/UTP templates, final push to final/main	.gitignore ignored Doc PDFs - fixed with !Doc/**/**.pdf	Verify git status PDFs tracked
2026-08-27	Sprint 6	Team Lead	Completed templates, final review, 10min + 5min Q&A dry-run	Submit to Classroom and GitHub by 28-Aug	None - submission ready	Upload and verify

CoachAI - Sprint Retrospectives (6 Sprints)

What went well / What went wrong / Improvements - reviewed at each sprint review

Sprint (Dates)	What Went Well	What Went Wrong / Challenges	Action Items / Improvements	Owner	Due Date / Status
Sprint 1 07-12 Jul 2026	Clear architecture and 9-step flow saved rework; Graceful degradation prevented late surprises; GitHub branching clean	Azure access delayed - started with Groq; Initial chunk 1024 too large, truncated tips	Request Azure key early; reduce chunk to 512 and sentence re-split; document 3-layer fallback	Team Lead AI Architect	12-Jul Done
Sprint 2 14-19 Jul 2026	<0.5s pipeline with ThreadPool 2 workers; Escalation formula easy to tune and test; Macros library made suggestions	LLM JSON failed when model added ""json fences; Clarity score variance across intents	Add regex JSON extractor + heuristic fallback for every agent; Tune coaching prompt with valid intents enum	AI Team QA Team	19-Jul Done
Sprint 3 21-26 Jul 2026	Hybrid RAG (semantic+BM25) works offline; KB-gap <0.45 + auto-KB draft delighted reviewers; Deep one-call cut latency 40%	OpenRouter 429 during demo; Viral/fraud keyword false negatives on Hinglish	Add retry + keyword seeded defaults + log; Expand Hinglish lists, tests for 'paise wapais'	AI Team	26-Jul Done
Sprint 4 28 Jul-02 Aug	25 agents single responsibility - prompts small/testable; Composio real actions + mock offline; Survival and Hall of Fame wow factor	Composio SDK breaking change 0.18->0.19; Manager whisper fired too often (>0.3); Survival singleton shared	Pin composio>=0.18.1,<0.20; Raise threshold to >=0.5 + hysteresis; Document singleton limit in gotchas	AI Team Backend Team	02-Aug Done
Sprint 5 04-09 Aug 2026	Node backend mirrors Python - frontend auth works (/api/auth/*); React 4-tab rail highly usable; Feature Lab strong demo	Two backends same Vercel project confusion; Risk tab overflow mobile; DB rehydration edge: cold instance lost _sessions	Deploy Node to coachai-backend, verify /health before frontend; Fix responsive; Add JSON rehydrate via full_state + activity log	Frontend Team Backend Team	09-Aug Done
Sprint 6 11-27 Aug 2026	Full suite green (pytest+vitest+node), p95 <0.8s; 10-slide PPT exactly + KT Guide + QA Bank ready; Ephemeral FS fix closed top bug	PPT initially 12 slides - had to trim; .gitignore ignored Doc PDFs - almost missed; Template filling took full day	Enforce slide count checklist; Add lDoc/**/**.pdf; Create generator script for next batch; Dry-run 27-Aug, submit 28-Aug	Team Lead QA Team All	27-Aug Done

Outcome: 30/30 user stories completed, 0 blockers pending, velocity ~5 stories/sprint. Top risks mitigated: two-backends-one-Vercel and ephemeral FS via pg JSONB rehydration and DB-authoritative reads. Next: unify to single canonical backend, add pgvector at scale.